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Subject: Programming Language Roll No.: 7 Exp. No.: \_\_\_\_\_  
Name of the Experiment: Introduction to Python language  
Performed on: \_\_\_\_\_  
Submitted on: \_\_\_\_\_

| Marks | Teacher's Signature with date |
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| 10    |                               |

Q1. Key Features of Python.

- Easy to Read and write: Python uses a simple, human-like syntax that relies on indentation rather than curly braces, making it highly readable & beginner friendly.
- Interpreted Language: Code is executed line by line by the interpreter, which simplifies debugging as errors are reported immediately during executing.
- Dynamically Typed: You don't need to declare variable types beforehand; Python assigns them automatically at runtime.

Key Advantages

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐

Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐

Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

Teacher's Signature

- High Productivity : The concise syntax allows developer to write less code to achieve the same result compared to language like C++ or Java, speeding up development
- Versatility : It supports multiple programming paradigms - including object-oriented, procedural, tutorial and functional programming allowing it to be used for almost any task
- Strong Community Support - A massive global community provide extensive documentation, tutorials, and third-party library that solved complex problem.

### Common Python IDEs & Editors

1. PyCharm : A full-featured, professional IDE developed by JetBrains that offers advanced tools for debuggings, code refactoring, and database integration
2. Visual Studio Code - A lightweight, open-source editor ~~for~~ from Microsoft that becomes a powerful Python environment through the use of extension